

CLASSIFIED

THE ANTIKYTHERA MECHANISM ANALYSIS

Recovered off Antikythera, Greece. 35.8617 N, 23.3067 E

CASE NO.:	HE-131-AK
FILE OPENED:	1901
STATUS:	UNRESOLVED - ACTIVE
DISTRIBUTION:	PURCHASER COPY - PERSONAL LICENSE

DO NOT CIRCULATE

CASE SUMMARY

In 1901, sponge divers pulled a corroded lump of bronze out of a Roman-era shipwreck. It sat in a museum crate while archaeologists catalogued the statues. When the lump split open, it showed gear teeth. Precision-cut bronze gearwheels, from a ship that sank around 60 BC.

The Antikythera Mechanism is a geared analog computer. It modeled the sun, the moon, the phases, eclipses decades in advance, and, per the latest reconstructions, the five known planets, in a bronze case the size of a shoebox. Nothing of comparable mechanical complexity appears in the surviving record for the next fourteen hundred years.

The question of record was stated by the man who studied it longest, Derek de Solla Price: finding this device in that wreck is like finding a jet plane in the tomb of Tutankhamun. One does not build a computer first. One builds it last, at the end of a long line of simpler machines. The line is missing.

This dossier assembles what the field admits, what the field buried, and the explanations the field refuses to put on the record. The reader is advised that every fact in the first category is documented, every silence in the second category is dated, and every theory in the third category fits the evidence better than the official story does.

TIMELINE OF RECORD

c. 60 BC	A cargo ship out of the eastern Aegean goes down off Antikythera carrying bronzes, glassware, coin, luxury goods bound for Rome, and one geared instrument.
1900	Sponge divers from Symi, blown off course, find the wreck at 45 meters. Captain Kontos reports a seabed of corpses, the statues, to authorities in Athens.
1901	Salvage season recovers the hoard. One diver dies, two are paralyzed. Among the finds, an unremarkable corroded lump, crated. FILE OPENED.
1902	Archaeologist Valerios Stais notices a gear wheel embedded in the split lump. The find is disputed immediately; the objection on record is that the object is too complex for the wreck date.

1903-1951	Forty-eight years of institutional silence. The most important machine ever recovered from antiquity sits in Athens essentially unstudied while the field publishes on pottery.
1958	Derek de Solla Price begins two decades of study, publishes Gears from the Greeks in 1974, establishing at least 30 gears and a differential arrangement, a design not otherwise seen until the modern era.
1976	Jacques Cousteau works the wreck, recovers coins fixing the sink date, reports the cargo field extends far beyond the excavated zone.
2005	X-ray microtomography reads the internal inscriptions, a built-in user manual naming eclipse cycles and games years. Tooth counts confirm the Metonic and Saros cycles in bronze.
2012-present	Return expeditions every season since 2014. New statues, a human skeleton, a bronze arm pointing to at least seven more statues in the sediment. Excavators state most of the cargo field is untouched.
2016	A second concentration of wreckage is confirmed on the seabed. Its cargo has never been fully catalogued in public.
2021	The UCL Antikythera Research Team publishes a full front-dial reconstruction with epicyclic planetary gearing. The count runs to some 69 gears in the complete device.
Present	Roughly a third of the mechanism survives. No second device, no workshop, no prototype and no fragment of a predecessor has ever been produced for public examination.



Fragment A, the largest surviving piece, National Archaeological Museum, Athens. The four-spoke main drive wheel is visible through the corrosion, along with the sandwiched gear plates behind it. This is what two thousand years of seawater leaves of a computer. The sea kept this unit out of the melting pot. Nothing kept the others.

Source: Photograph by Giovanni Dall'Orto, 2009. National Archaeological Museum, Athens. Used with attribution per license.

THE OFFICIAL STORY

The consensus attributes the mechanism to Hellenistic craftsmen, probably of the school of Hipparchus or later Rhodes workshops, second or first century BC. Cicero mentions geared planetaria associated with Archimedes and Posidonius, so the capability is said to be documented. The device is called the culmination of a Greek mechanical tradition.

A culmination with no line to culminate. The tradition that supposedly produced it left no other gearwork of remotely comparable complexity, no workshop debris, no apprentice pieces, no failed castings, nothing. The literary references are one-line remarks about wonders owned by famous men, exactly the shape a rumor takes. The physical record contains this one object, alone.

Understand what the official story is asking you to accept. A civilization builds one computer, from a standing start, gets it right, and then never builds anything geared again for fourteen hundred years. No drafts. No successors. No copies. In the entire history of technology, no other invention has ever behaved this way. Inventions leave trails. This one left a hole.

WHAT THE RECORD SHOWS

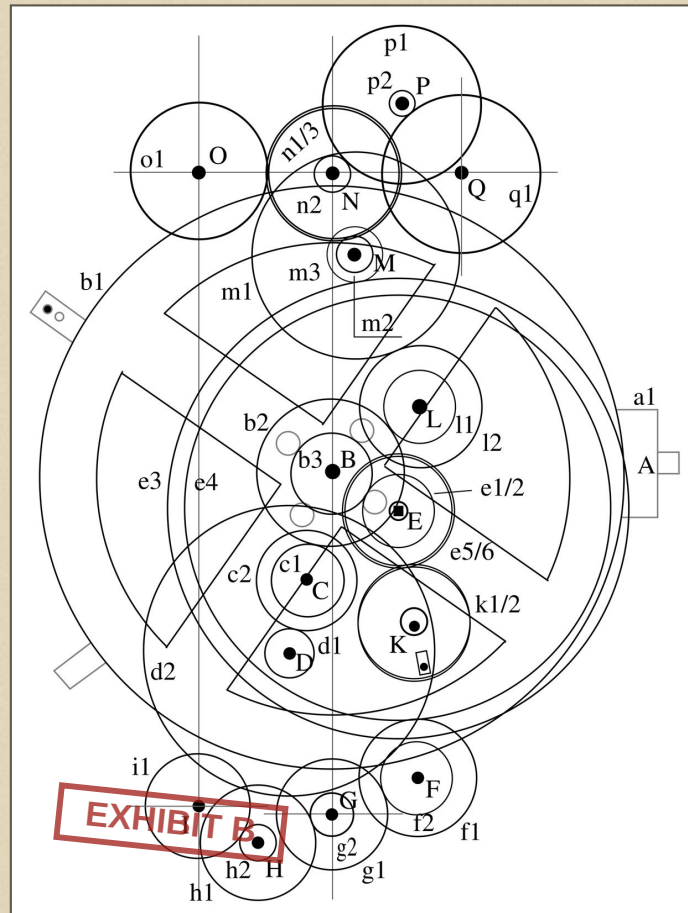
The engineering, first. The surviving gears are cut from bronze sheet around two millimeters thick, teeth triangular, some wheels carrying 223 teeth, a prime number chosen because the Saros eclipse cycle is 223 lunar months. A pin-and-slot arrangement rides one wheel on another to reproduce the moon's variable speed, which means the maker mechanically encoded lunar anomaly, an effect not formally theorized until Kepler. The tooth counts are not decoration. They are astronomy, compiled.

The device knew things its era is not supposed to have known this precisely. The inscriptions carry eclipse predictions with times of day and color forecasts for the eclipsed moon. The 2021 reconstruction gives the planets their correct synodic periods through compound gear trains accurate to a day in decades. This is not a model in the display sense. It is computation, executed in metal.

Second, the tolerances. The gear teeth are cut to a precision of a fraction of a millimeter, by hand, in bronze, and the axle holes are positioned to accuracies that modern machinists respect. Researchers who have tried to replicate the mechanism with period tools report that the hardest problem is not the design. It is the manufacturing. The maker had solved problems of precision fabrication that the official timeline says nobody had even encountered yet.

Third, the isolation. Price wrote that the mechanism forces a complete rethink of Greek technology, and then spent twenty years watching the field decline to rethink it. The standard histories still date geared clockwork to medieval Europe. The one surviving counterexample is handled as a marvel, an outlier, a [REDACTED] vanished]] except for the unit that happened to sink in deep water where no one could melt it down.

That last clause is the uncomfortable one. Bronze was money. Every geared device that stayed on land was eventually scrap. The only reason this file exists is that the sea kept one unit out of reach. The correct inference from one survivor in an environment that destroys survivors is not that one was made. It is that the others did not survive. The wreck did not preserve an exception. It preserved the only sample.



Reconstructed gear-train schematic. Every axle and wheel shown here is attested in the surviving fragments or required by the tooth counts. Metonic dial, Saros eclipse dial, Exeligmos corrector, lunar anomaly pin-and-slot, all driven from one hand crank. This is the anatomy of the machine the textbooks call an isolated curiosity.

Source: Gear-train schematic released CC0 via Wikimedia Commons.

THE FIFTY-YEAR BURIAL

Between 1902 and 1951, the most consequential artifact ever recovered from the ancient world received almost no formal study. Reflect on that span. Two world wars were fought, powered flight went from Kitty Hawk to jet engines, and the one object proving the ancient world had computing

machinery sat in a drawer in Athens.

The field's own record explains why. The first scholars to examine it said openly that the object could not be what it appeared to be, because Greeks of 60 BC did not have this technology. The premise was protected and the evidence was shelved. When German philologist Albert Rehm worked out around 1905 that the device was an astronomical calculator, his conclusions went unpublished in his lifetime. His notes, which had the answer forty years early, [REDACTED] century later]].

This is the pattern this office flags across every case in the series. Evidence that fits the timeline gets published in a season. Evidence that breaks the timeline gets a drawer. The drawer is not an accident. The drawer is the mechanism by which the timeline defends itself.

Ask the obvious question. If Rehm had the answer in 1905, and Price confirmed it by 1974, why do schoolbooks printed this year still teach that mechanical computation begins in medieval Europe? A fifty-year delay can be called neglect. A hundred-and-twenty-year delay is a decision.

UNSANCTIONED EXPLANATIONS

The official account, a lone genius workshop with no inputs and no outputs, is the least probable explanation on the table. This office records the alternatives that fit the evidence and notes that none of them has ever been refuted. They have only been refused.

THEORY ONE. THE INHERITANCE. The mechanism is a late copy of far older technology, preserved and transmitted from a civilization erased before recorded history. On this reading the Greeks did not invent the device. They maintained it, the way medieval monks copied Roman texts they could no longer fully understand. This is precisely how the mechanism behaves in the record: it appears complete, without development, the signature of a technology received rather than invented. Plato wrote down the Egyptian priests' warning to Solon that the Greeks were children who remembered nothing of the ages before the deluges. The priests said there had been many destructions of mankind. A device inherited across such a destruction would look exactly like this one.

THEORY TWO. THE ALEXANDRIA ERASURE. The developmental chain existed, in full, on paper, and the paper burned. The Library of Alexandria held the engineering corpus of the ancient Mediterranean, and it was destroyed in stages, by fire, by decree, by mob, until nothing remained. Archimedes'

treatise On Sphere-Making, which ancient authors say described the construction of exactly this kind of device, survives in zero copies. Not one. The only major work of the ancient world's greatest engineer that is completely lost is the one that explained the machines. Draw your own conclusion about which books get lost.

THEORY THREE. THE CLOSED GUILD. The gearwork tradition was real, generations deep, and deliberately secret. Precision astronomical machinery in the ancient world was power: calendars ruled festivals, eclipses ruled omens, and omens ruled politics. A guild that could predict an eclipse to the hour owned something worth killing for, and guilds that own such things do not publish. They transmit by apprenticeship, they encode, and when the line breaks, everything vanishes at once. The record shows exactly this shape: total silence, one leaked unit, total silence.

THEORY FOUR. THE WRONG DATE. A minority of researchers has argued from the eclipse data that the mechanism's astronomical epoch points to 205 BC, and others note the device's style matches nothing else from any known workshop of any known century. If the object is centuries older than the wreck that carried it, it was already an antique when it sank, a relic being transported, [REDACTED], from a source the shippers themselves may not have understood. The cargo manifest of that ship is the single most important lost document in the history of technology.

THEORY FIVE. CONTACT. The boldest reading on file: the astronomical knowledge encoded in the gearing, lunar anomaly, planetary periods, eclipse color prediction, was not developed by observation at all but received, whole, from an intelligence that already had it. The ancient world's own literature supports the shape of this claim. The Babylonians, whose eclipse cycles the mechanism runs on, said their sciences were gifts from beings who came from the sky before the flood. The field calls that mythology. The field also cannot explain where the Babylonian eclipse data came from, because the observation baseline required is longer than the civilization that held it.

ANOMALIES ON FILE

ITEM 1. Complexity without ancestry. Some 69 gears, differential and epicyclic arrangements, in a record that otherwise contains no Greek gearwork more advanced than a windlass. The developmental chain is one hundred percent missing.

ITEM 2. Complexity without descendants. The skills evaporate for fourteen centuries, until European astronomical clocks, which arrive, again, near fully formed, in the same corner of the world the wreck's cargo was sailing toward.

ITEM 3. The maker encoded lunar anomaly mechanically eighteen centuries before the mathematics existed to describe it formally.

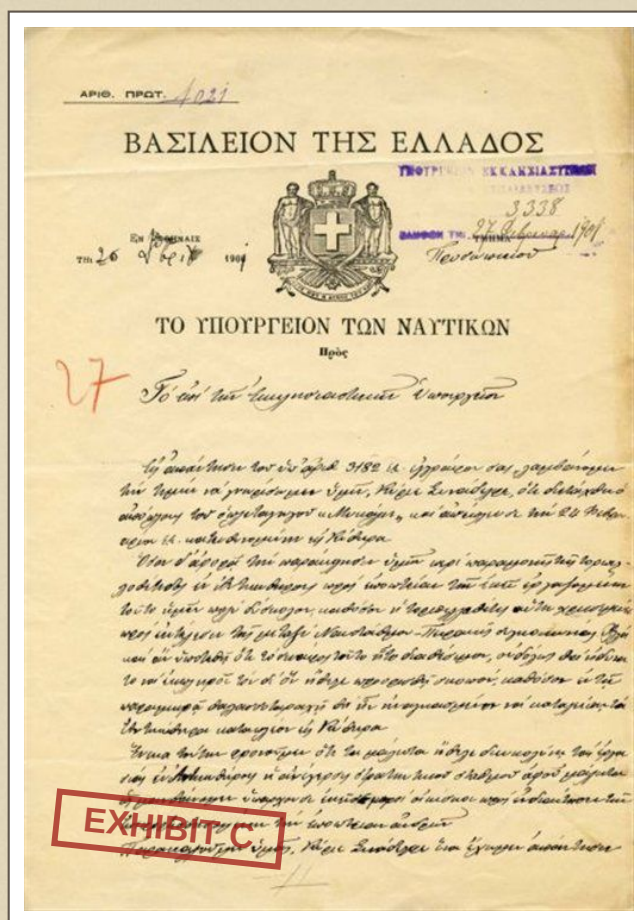
ITEM 4. The initial 1902 identification was resisted on the explicit grounds that the object was too advanced for its date. The date won. The implications were then filed under craftsmanship and left there.

ITEM 5. Albert Rehm solved the device's function around 1905. His work was never published in his lifetime. The solution sat in an archive for a century while the public was taught the ancients had no machines.

ITEM 6. Archimedes' On Sphere-Making, the one ancient text describing construction of geared planetaria, is lost in its entirety. Every other major Archimedean work survives in some form. The engineering manual alone is gone.

ITEM 7. The wreck lies on the direct luxury route to Rome, and Cicero, writing within a decade of the sinking, describes two geared spheres in Roman hands, both taken from Syracuse as war plunder a century and a half earlier. Rome collected these devices. Rome never learned to make them. Whoever made them was already gone.

ITEM 8. Expeditions returned in 1953, 1976, 2012 and every season since 2014. The wreck is still producing material, and the excavators state on record that most of the cargo field is untouched. If a second mechanism exists anywhere, it is there, and every season it stays on the seabed is another season the file stays open.



Hellenic Ministry of Naval Affairs dispatch, 26 February 1901, ordering the vessel Mykali to the Antikythera wreck for the salvage season. The paperwork that pulled the mechanism off the seabed. The government filed the recovery. The field filed the implications, and never retrieved them.

Source: Hellenic Ministry of Naval Affairs, 1901. Public domain.

QUESTIONS THE FIELD WILL NOT PUT IN WRITING

Where are the other units? Cicero saw two in Rome. The inscriptions on the recovered device read as a user manual, and nobody writes a user manual for a one-off. Manuals mean production. Production means a shop, inventory, customers. Every one of those objects went somewhere, and the somewhere has never been searched because the field does not want to find it.

Why does the recovered device carry no maker's mark, no dedication, no owner's name? Ancient craftsmen signed far lesser works. An object of this value, anonymous, is an object whose origin was either unknown to its own era or deliberately unstated.

Who funded the fifty-year silence? Museums answer to boards, boards answer to donors, and the history of archaeology is a history of inconvenient objects described once and never exhibited. The mechanism spent decades as a crate number while lesser bronzes from the same wreck toured the world. Somebody decided the statues were the story. This office wants the name.

And the question under all of it: how many other drawers are there? The mechanism was found by accident, by divers hunting sponges, in one wreck, on one route. The Mediterranean holds thousands of unexcavated wrecks. The technology existed. The shipping lanes existed. The mathematics says the seabed is holding more, and the funding to look is [REDACTED] to safer digs]].

FINAL ASSESSMENT

The Antikythera Mechanism is treated as a delightful exception, and exceptions are where the record admits what it otherwise conceals. A civilization does not produce one computer. The device testifies to a mature engineering lineage, generations deep, whose every other product was recycled into coins and cannon metal, and whose existence was compressed by history into two sentences of Cicero.

This office assesses the mechanism as recovered property of a lost technical tradition, scale unknown, duration unknown, origin unresolved between the five theories on file, and notes that the knowledge it embodied had to come from somewhere and go somewhere. Fourteen hundred years is not a gap. It is an erasure, and erasures have authors.

The cargo field at 45 meters is still being worked. The second concentration of wreckage has not been fully catalogued in public. Rehm's archive proved the answers can sit in drawers for a century. The file holds position until the sea, or the drawers, give up the second unit.

CASE REMAINS OPEN